Tanveer Ahmad

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EDUCATION

COMSATS University Islamabad, Islamabad Campus
Bachelors of Software Engineering — *Sep 2015 - May 2019*

EXPERIENCE

Principal Software Engineer — May 2023 - Present
Arbisoft — *Islamabad, PK*

Worked on multiple projects focused on building automation systems and AI-driven solutions to improve product quality, engineering efficiency, and platform intelligence. Contributed to backend development, AI integration, testing automation, and system reliability across different teams.

- Built an in-house automation platform to streamline QA workflows and reduce manual effort
- Developed a Quality Dashboard for real-time test and quality metrics
- Automated release notes, Slack notifications, and JIRA ticket creation for regression testing
- Designed and maintained Pytest + Azure DevOps CI/CD test automation framework
- Integrated Gemini APIs into AI-driven document intelligence architecture
- Built and maintained FastAPI-based REST services for backend and AI workflows
- Developed an AI recruitment platform: candidate matching, LiveKit-based AI voice interviews, and OpenAI evaluation reports
- Built async background workers (TaskIQ + Redis) and API observability with request logging, redaction, and scheduled cleanup
- Used Claude Code with custom skills, commands, and hooks to speed up development, code review, and test generation
- Supported onboarding and mentoring of new engineers

Senior Software Engineer. — Mar 2022 - Apr 2023
Emumba — *Islamabad, PK*

Working on a SaaS-based platform for cryptocurrency tax automation and real-time portfolio analytics, focusing on building scalable backend systems and seamless service integration.

Responsibilities:

- Built and maintained scalable backend systems using Python and Django REST Framework
- Integrated crypto wallets and third-party APIs for automated data ingestion using Airflow
- Collaborated with cross-functional teams to ensure smooth feature delivery
- Managed source control with GitLab and handled database operations using PostgreSQL
- Conducted code reviews and contributed to system design and new feature development
- Deployed and maintained backend services on AWS for scalability and security

Key Achievements:

- Enabled automated crypto data ingestion for accurate tax reporting
- Improved system scalability and reliability through robust backend architecture
- Streamlined developer collaboration with well-structured APIs and documentation

Software Engineer — Apr 2021 - Jan 2022
Afiniti — *Islamabad, PK*

Worked on a Text-to-Speech (TTS) engine for the Mega product, enabling natural-sounding speech synthesis across multiple voices, languages, and formats.

Responsibilities:

- Developed REST APIs for the TTS engine using FastAPI
- Integrated trained speech synthesis models with backend services
- Enabled multi-language and multi-voice speech generation
- Maintained and optimized the existing TTS engine, including bug fixes
- Mentored and guided junior team members
- Ensured clean, readable, and standardized code following PEP 8

Key Achievements:

- Delivered a scalable TTS solution supporting multiple languages and voice variants
- Improved system reliability through continuous maintenance and optimization
- Contributed to efficient team collaboration and knowledge sharing

Team Lead / Full-stack Developer Nov 2019 - Apr 2021
Rapidev DMCC *Islamabad, PK*

Led and contributed to the development of scalable web automation and data platforms, focusing on backend architecture, data ingestion, and full-stack feature delivery.

Responsibilities:

- Designed and implemented distributed web crawling architecture for large-scale data ingestion
- Developed microservices using Flask, Celery, and RabbitMQ
- Built web crawlers using Scrapy and Selenium for data extraction and normalization
- Created backend APIs and a crawler management dashboard using Django REST Framework
- Implemented anti-scraping and reverse engineering techniques for complex data sources
- Led Agile processes including sprint planning, standups, and retrospectives
- Managed task allocation, conducted code reviews, and ensured high-quality delivery
- Mentored 10+ engineers and established onboarding SOPs
- Conducted 200+ technical interviews to scale the engineering team

Key Achievements:

- Built and scaled a high-performance data crawling and automation system
- Achieved high reliability in data extraction across multiple platforms
- Improved team productivity and contributed to rapid engineering team growth

PROJECTS

BenchPrep | *Python, FastAPI, SQLAlchemy, Alembic, Pydantic, PostgreSQL, Redis, TaskIQ, OpenA* Apr 2026 - Present
AI recruitment platform with AI voice interviews

- Built REST APIs for candidates, roles, interviews, and dashboards using FastAPI,
- SQLAlchemy 2.0 async and Pydantic V2.
- Developed the LiveKit AI voice interview agent — turn-taking, silence handling,
- and error fallbacks.
- Implemented the interview evaluation pipeline using OpenAI and Whisper: transcription, speech metrics, and AI-generated reports.
- Built API request logging with async queuing, batched inserts, data redaction, and a scheduled TaskIQ cleanup job.
- Wrote Alembic migrations and pytest test coverage for all changes.
- Used Claude Code to speed up development — custom skills, slash commands, and hooks for automated code review, test generation, and repo-specific workflows.

Docugami | *FastAPI, Python, Pandas, Gemini API, GCP, Zapier, Microsoft Power Automate* Dec 2023 - Apr 2026
AI-based solutions for document processing

- Integrating Gemini APIs into the solution architecture.
- Designing and refining prompts for various features.
- Developing RESTful APIs using FastAPI.
- Utilizing Pandas and other Python libraries for data manipulation and analysis.
- Deploying the solution on Google Cloud Platform (GCP).
- Onboarding new team members and helping them get up to speed.
- Building workflows with Zapier and Microsoft Power Automate

Kayak - the Gatherer | *Python, JIRA & SLACK APIs, Atlassian Bamboo, Microsoft Azure, Azure DevOps* May 2023 - Nov 2023
QA dashboard with Auto release notes & JIRA ticket

- Developed internal solutions to optimize workflows and client needs
- Created a real-time Quality Dashboard for monitoring metrics
- Automated Release Notes generation and JIRA ticket creation
- Integrated Slack notifications for better team communication
- Implemented Pytest pipeline on Azure DevOps for end-to-end testing

Capital-Bee (FinTech application) | *Python, Django REST Framework, PostgreSQL, Airflow, GitLab, AWS* Mar 2022 - Apr 2023
SaaS-based solution that automates tax filing

1. Built scalable fintech portal backend systems using Django REST Framework
2. Integrated crypto wallets and automated data ingestion via Airflow and APIs
3. Maintained API documentation for frontend collaboration
4. Managed GitLab source control and PostgreSQL databases
5. Conducted code reviews and contributed to feature design
6. Deployed backend services to AWS for secure, scalable production

Text To Speech engine | *Python, FastAPI, Django, PostgreSQL, GIT* Apr 2021 - Jan 2022

TTS engine with multiple voices, languages

- API Development for TTS engine using Fast API
- Integration of well-trained models with REST Server (TTS Engine).
- Daily standup and meeting with the team members
- Maintenance of existing TTS engine and fixing bugs.
- Collaborating and guiding junior/new resources.

OWLSENSE | *DRF, Flask, Celery, RabbitMQ, Scrapy, Selenium, Microservices, Web scraping* Nov 2019 - Apr 2021

OSINT-driven data platform

- Built microservices using Flask, Celery, and RabbitMQ
- Developed web crawlers using Scrapy and Selenium for data extraction
- Implemented anti-scraping and reverse engineering techniques
- Created crawler management dashboard using Django REST Framework
- Designed scalable backend APIs and data processing pipelines
- Led distributed web crawling architecture for large-scale data ingestion
- Owned end-to-end feature delivery, from requirements to deployment
- Managed task assignment, code reviews, and ensured delivery quality
- Mentored 10+ engineers and defined onboarding SOPs
- Conducted 200+ technical interviews to scale the team

PlantCare | *ReactJS, Django REST Framework, Python, TensorFlow, Keras* May 2019 - Nov 2019

From leaf image to diagnosis in seconds

- Built REST APIs using Django for ML model integration
- Developed ReactJS frontend portal
- Integrated CNN model with backend for real-time predictions
- Handled image processing and API data flow
- Trained and optimized deep learning model for disease detection